

Module 2: Transport Layer and Network Layer



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Transport Layer

- Services and Protocols
- UDP and TCP

Hands-on

- Sockets Introduction
- Elementary TCP Sockets
- TCP Client/Server Example
- I/O Multiplexing
 - select()
 - poll()
- Elementary UDP Sockets
- Advanced I/O Functions

Network Layer

- Introduction
- Network-layer Protocols
- Unicast Routing
- Multicast Routing
- Multicasting Basics
- Intra-domain Routing
- Inter-domain Routing
- Next Generation IP (IPv6)
- Quality of Service (QoS)

Hands-on

- Linux Routing Table
- Routing Caches
- Routing Cache Implementation
- Adding Routes using ip

4.5 Next Generation IP – IPv6

- IPv6 (Internet Protocol version 6), also called IPng, was proposed in the early 1990s.
- Main motivation: depletion of the IPv4 address space and other shortcomings.
- IPv6 enlarges the address space and redesigns the IP packet format.
- Auxiliary protocols such as ICMP were also revised.
- IPv5 was an earlier proposal based on the OSI model that never materialized.

Main Changes in IPv6

Address and Header

- 128-bit address space
- Better header format
- Options separated from base header
- New options

Services and Extension

- Designed for future extension
- Traffic class and flow label
- Support for resource allocation
- Encryption and authentication support

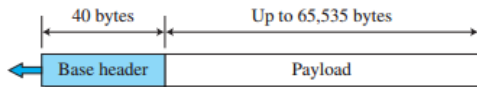
Why IPv6 Adoption Was Slow

- The original motivation—IPv4 address depletion—was slowed by short-term remedies:
 - Classless addressing
 - DHCP for dynamic address allocation
 - NAT
- Continued Internet growth and services such as mobile IP, IP telephony, and IP-capable mobile telephony create pressure for IPv6.
- IPv6 adoption therefore requires a gradual transition rather than a sudden replacement.

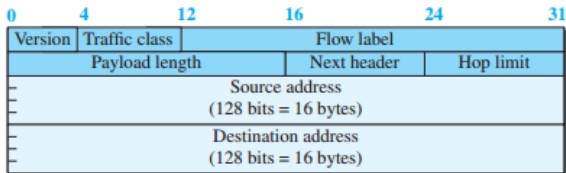
- An IPv6 packet consists of a **40-byte base header** followed by the payload.
- Payload can contain up to **65,535 bytes** of information.
- The base header has a fixed length.
- Options are carried using **extension headers**.

Figure 4.101 – IPv6 Packet Format

Figure 4.101 *IPv6 datagram*



a. IPv6 packet



b. Base header

IPv6 Base Header – Fields I

Version

- 4-bit field
- Value is 6 for IPv6

Traffic Class

- 8-bit field
- Distinguishes payloads with different delivery requirements
- Replaces IPv4 type-of-service field

Flow Label

- 20-bit field
- Identifies a particular flow
- Supports special handling

Payload Length

- 2-byte field
- Length of datagram excluding the base header
- Base header length is fixed at 40 bytes

Next Header

- 8-bit field
- Identifies the first extension header or the carried protocol
- Similar to IPv4 protocol field

Hop Limit

- 8-bit field
- Same purpose as IPv4 TTL

Source Address

- 16 bytes = 128 bits
- Identifies the original source

Destination Address

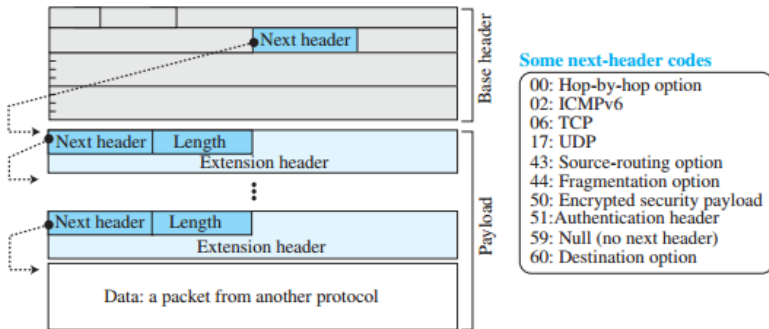
- 16 bytes = 128 bits
- Identifies the destination

IPv6 Payload and Extension Headers

- IPv6 payload consists of:
 - 1 Zero or more extension headers
 - 2 Data from protocols such as UDP or TCP
- IPv4 options are redesigned as extension headers in IPv6.
- Any number of extension headers can be included as required.
- Each extension header contains:
 - Next Header
 - Length
 - Option-specific information
- The last Next Header identifies the carried protocol such as UDP or TCP.

Figure 4.102 – IPv6 Payload and Extension Headers

Figure 4.102 *Payload in an IPv6 datagram*



Flow and Priority in IPv6

- IP was originally designed as a connectionless protocol.
- IPv6 introduces a **flow label** directly into the datagram format.
- A **flow** is a sequence of packets sharing characteristics such as:
 - Same path
 - Same resources
 - Same security requirements
- A router supporting flow labels maintains a **flow label table**.

Flow Label Table

- Each active flow label has an entry describing the required services.
- When a packet arrives:
 - ① Router reads the flow label.
 - ② Router consults the flow label table.
 - ③ Corresponding services are provided to the packet.
- The flow label itself does not contain all service information.
- Other mechanisms, such as hop-by-hop options or other protocols, provide the required information.

Uses of the Flow Label

Simple Form

- Speeds up router processing.
- Router can consult a flow-label table for the next hop.
- Avoids repeatedly consulting the forwarding table and routing algorithm.

More Sophisticated Form

- Supports real-time audio and video.
- Such traffic may need high bandwidth, large buffers, and processing resources.
- Resources may be reserved in advance.

RTP and RSVP may be used together with IPv6 for real-time transmission and resource reservation.

Fragmentation and Reassembly in IPv6

- IPv6 still supports fragmentation and reassembly.
- **Major difference:** routers do not fragment IPv6 datagrams.
- Fragmentation is performed only by the **source**.
- Reassembly takes place at the **destination**.
- This reduces processing at routers.

IPv6 Fragmentation – Operation

- ➊ Source checks the packet size and decides whether fragmentation is needed.
- ➋ Router checks the packet size against the next network's MTU.
- ➌ If the packet is too large, the router drops it.
- ➍ Router sends an **ICMPv6 Packet Too Big** error message to the source.
- ➎ Source can then adjust the packet size.

IPv6 Extension Headers

- Extension headers are an important part of IPv6.
- Examples discussed in the text include:
 - Fragmentation header
 - Authentication header
 - Extended Security Payload header
- They provide additional functionality without enlarging the fixed base header.

4.5.2 IPv6 Addressing

- IPv6 addresses are **128 bits (16 bytes)** long.
- This is four times the length of an IPv4 address.
- Human-readable representation is therefore necessary.
- Two representations discussed:
 - Binary notation
 - Colon hexadecimal notation

IPv6 Colon-Hexadecimal Notation

- Address is divided into **eight sections**.
- Each section contains four hexadecimal digits.
- Sections are separated by colons.

FEF6:BA98:7654:3210:ADEF:BBFF:2922:FF00

Leading-Zero Abbreviation

- Leading zeros in a section may be omitted.
- Example: 0074 → 74, 000F → F, 0000 → 0.

IPv6 Zero Compression

- Consecutive sections containing only zeros can be replaced by ::.
- Zero compression can be used **only once** in an IPv6 address.

FDEC:0:0:0:0:BBFF:0:FFFF

↓
FDEC::BBFF:0:FFFF

Rule

If more than one run of zero sections exists, only one run can be compressed.

Mixed IPv6 Representation

- During IPv4-to-IPv6 transition, an IPv4 address may be embedded in the rightmost 32 bits.
- Leftmost six sections use colon hexadecimal notation.
- Rightmost two sections can use dotted-decimal notation.

::130.24.24.18

- The zero compression indicates that the leftmost 96 bits are zero.

CIDR Notation in IPv6

- IPv6 supports hierarchical addressing.
- Therefore, slash/CIDR notation can be used.

FDEC::BBFF:0:FFFF/60

Meaning

The /60 indicates that the first 60 bits form the prefix.

IPv6 Address Space

- IPv6 has an address space of:

$$2^{128}$$

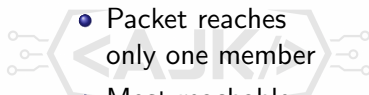
- This is 2^{96} times the IPv4 address space.
- Even if only 1/64 of the address space were assigned to people on Earth, the number of addresses per person would remain enormous.
- The text illustrates the scale as approximately 2^{88} addresses per person under its assumptions.

Three IPv6 Address Types

Unicast

- One interface
- One-to-one communication

Anycast

- Group of computers share an address
 - Packet reaches only one member
 - Most reachable member
- 
- A diagram illustrating anycast. It features a central hexagonal node with the word 'ANCA' inside. To the left of the hexagon, three lines with small circles at their ends converge towards the node. To the right, three lines with small circles at their ends also converge towards the node, representing multiple sources or destinations sharing a single address.

Multicast

- Group of computers
- Every member receives a copy

Broadcasting

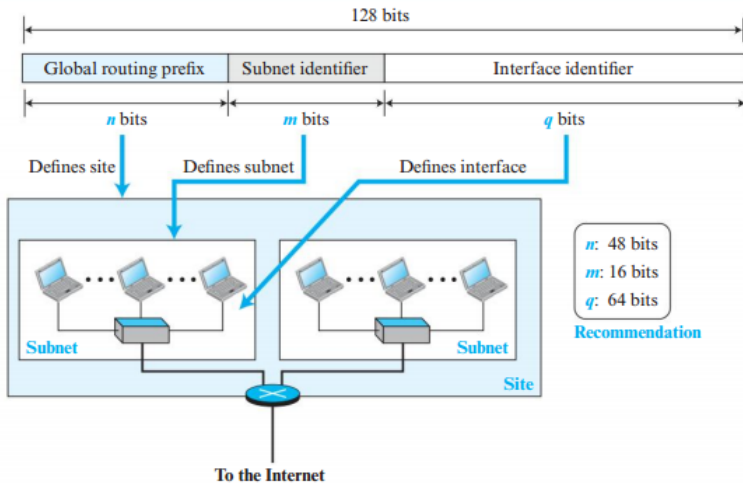
IPv6 does not define a separate broadcasting address type; broadcasting is treated as a special case of multicasting.

IPv6 Address Space Allocation

Prefix	CIDR	Assignment	Fraction
0000 0000	0000::/8	Special addresses	1/256
001	2000::/3	Global unicast	1/8
1111 110	FC00::/7	Unique local unicast	1/128
1111 1110 10	FE80::/10	Link local	1/1024
1111 1111	FF00::/8	Multicast	1/256

Figure 4.103 – Global Unicast Address Structure

Figure 4.103 Global unicast address



Global Unicast Addresses

- Global unicast block: 2000::/3.
- The first three bits are fixed as 001.
- An address is divided into:
 - 1 Global routing prefix
 - 2 Subnet identifier
 - 3 Interface identifier

Global Unicast Address – Components

Global Routing Prefix

- Routes packets through the Internet to an organization/site.
- Remaining 45 bits can identify up to 2^{45} sites under the recommended structure.

Subnet Identifier

- Recommended 16 bits.
- Up to $2^{16} = 65,536$ subnets.

Interface Identifier

- Recommended 64 bits.
- Identifies an interface rather than the host itself.

Link-Layer Relationship

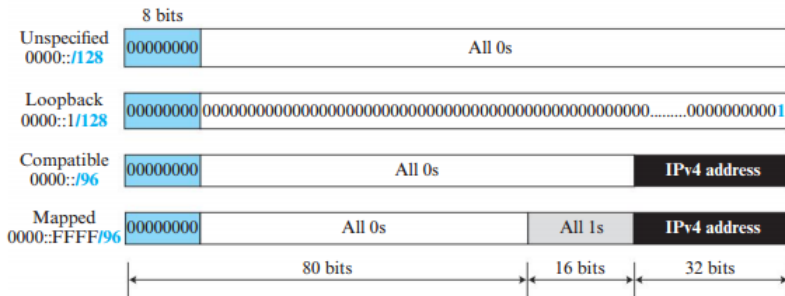
- EUI-64 and Ethernet link-layer addresses can be related to the interface identifier.

Special IPv6 Addresses

- The 0000::/8 block is reserved; part is used for special addresses.
- **Unspecified address:** one address used during bootstrap when a host does not know its own address.
- **Loopback address:** one address used for loopback communication.
- IPv6 also defines IPv4-compatible and IPv4-mapped address formats for transition.

Figure 4.104 – Special IPv6 Addresses

Figure 4.104 *Special addresses*



IPv4-Compatible and IPv4-Mapped Addresses

Compatible Address

- 96 zero bits followed by 32-bit IPv4 address.
- Used when an IPv6 computer communicates with another IPv6 computer using an embedded IPv4 address.

Mapped Address

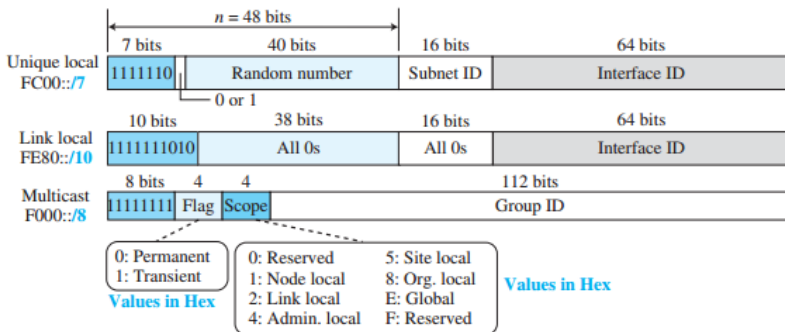
- Used when an IPv6 system sends toward a system still using IPv4.
- Supports transition between versions.

Checksum Observation

The text notes that the extra zero/one groups do not affect checksum calculation in the relevant pseudoheader context.

Figure 4.105 – Other Assigned IPv6 Blocks

Figure 4.105 *Unique local unicast block*



Other Assigned IPv6 Blocks

Unique Local Unicast

- Intended for private site use.
- Not expected to be routed on the public Internet.
- Uses prefix FC00::/7.
- Includes a 40-bit randomly generated identifier in the described structure.

Link Local

- Uses block identifier 1111111010.
- Next 54 bits are zero.
- Last 64 bits identify the interface.

IPv6 Multicast Addresses

- IPv6 assigns a large block for multicast.
- All multicast addresses use the prefix:

11111111

- The second field is a **flag**:
 - Permanent group
 - Transient group
- The third field defines the **scope** of the group.
- Scope values support different ranges of multicast operation.

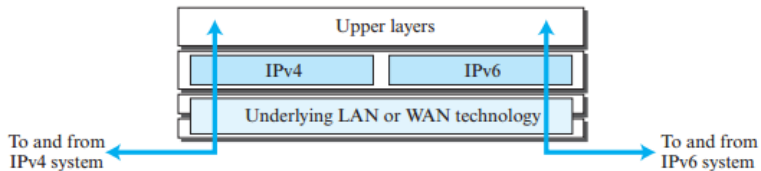
4.5.3 Transition from IPv4 to IPv6

- A sudden transition day is impractical because of the huge number of Internet systems.
- IPv4 and IPv6 must coexist during a gradual transition.
- The IETF recommended three strategies:
 - ① **Dual stack**
 - ② **Tunneling**
 - ③ **Header translation**

- A host runs IPv4 and IPv6 simultaneously during transition.
- Before complete migration, the station maintains both protocol stacks.
- DNS helps determine which version to use:
 - DNS returns IPv4 address → send IPv4 packet
 - DNS returns IPv6 address → send IPv6 packet

Figure 4.106 – Dual Stack

Figure 4.106 *Dual stack*

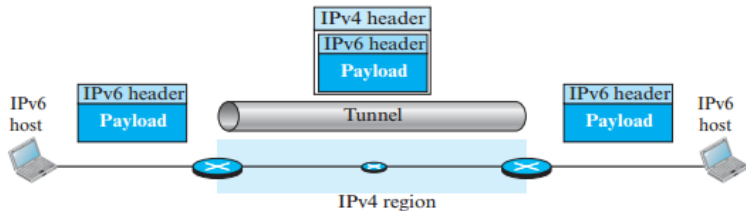


Tunneling

- Used when two IPv6 systems communicate across an IPv4 region.
- The IPv6 packet is encapsulated inside an IPv4 packet when entering the IPv4 region.
- It is decapsulated when leaving the region.
- The IPv4 protocol value is set to **41** to indicate that the IPv4 packet carries IPv6 data.

Figure 4.107 – IPv6 Tunneling

Figure 4.107 *Tunneling strategy*

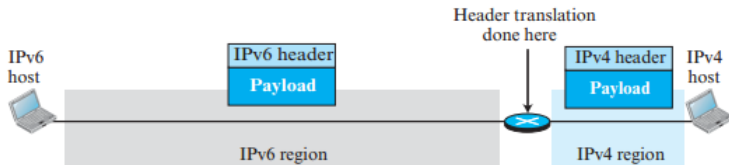


Header Translation

- Used when most of the Internet has moved to IPv6 but some receivers still use IPv4.
- Tunneling cannot solve this case because the receiver must understand an IPv4 packet.
- The IPv6 header is therefore converted into an IPv4 header.
- This allows communication between IPv6 and IPv4 systems.

Figure 4.108 – Header Translation

Figure 4.108 *Header translation strategy*



IPv4-to-IPv6 Transition – Comparison

Strategy	Situation	Main Operation
Dual stack	Hosts can support both versions	Run IPv4 and IPv6 simultaneously
Tunneling	IPv6 traffic crosses an IPv4 region	Encapsulate IPv6 inside IPv4
Header translation	IPv6 sender communicates with IPv4-only receiver	Convert IPv6 header to IPv4 header

4.5.4 ICMPv6

- ICMPv6 is the IPv6 version of the Internet Control Message Protocol.
- It follows the general purposes of ICMPv4 but is more extensive.
- Some protocols that were independent in IPv4 are incorporated into ICMPv6.
- New message types are added.
- In the IPv6 network layer, ICMP, ARP, and IGMP functions from IPv4 are combined into ICMPv6.

Figure 4.109 – IPv4 and IPv6 Network Layer

Figure 4.109 *Comparison of network layer in version 4 and version 6*



1. Error-Reporting Messages

- Report problems in packet delivery or processing.

2. Informational Messages

- Correspond to query-type functions.
- Echo request
- Echo reply

3. Neighbor-Discovery Messages

- Discover routers
- Discover link-layer addresses
- Discover IPv6 addresses
- Redirect messages

4. Group-Membership Messages

- Support multicast group membership.

Informational Messages in ICMPv6

- The informational group has one pair of messages:
 - Echo-request
 - Echo-reply
- They play a role similar to query messages in IPv4.
- Timestamp request and timestamp reply were dropped because they were rarely used.

Neighbor Discovery in ICMPv6

- Neighbor-discovery messages are a new group in IPv6.
- They can:
 - Find routers in the neighborhood
 - Find link-layer addresses of neighboring hosts
 - Find IPv6 addresses of neighboring hosts
 - Redirect messages

Neighbor Discovery – Message Types

Router Discovery

- Router Solicitation
- Router Advertisement
- Helps a host find a default router.

Neighbor Address Resolution

- Neighbor Solicitation
- Neighbor Advertisement
- Finds link-layer address from IPv6 address.

Redirection

- Performs the redirection function.

Inverse Neighbor Discovery

- Inverse Neighbor Solicitation
- Inverse Neighbor Advertisement
- Finds IPv6 address from a link-layer address.

IPv6 – Section Summary

- IPv6 provides a 128-bit address space.
- The base header is fixed at 40 bytes.
- Traffic class and flow label support differentiated handling.
- Extension headers provide optional functionality.
- IPv6 fragmentation is performed only by the source.
- IPv6 supports unicast, anycast, and multicast.
- IPv4-to-IPv6 transition uses dual stack, tunneling, and header translation.
- ICMPv6 combines and extends several IPv4 network-layer control functions.

Key Exam Points

- IPv6 address length: **128 bits**
- IPv6 base header: **40 bytes**
- Flow label: **20 bits**
- Traffic class: **8 bits**
- Hop limit: **8 bits**
- Global unicast prefix: 2000::/3
- Multicast prefix: FF00::/8
- IPv6 fragmentation: **source only**
- IPv4-over-IPv6 transition strategies: **dual stack, tunneling, header translation**